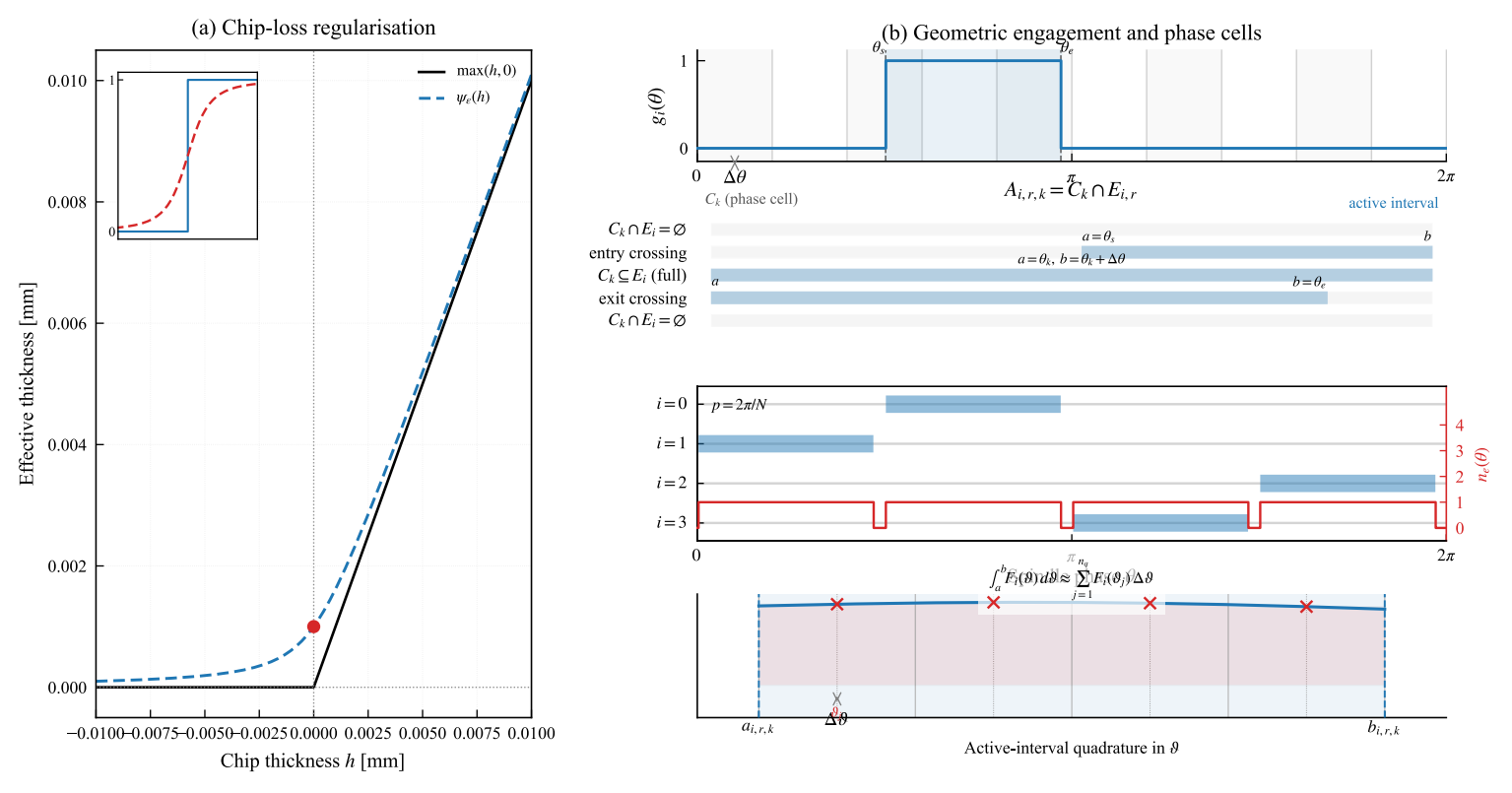

(a) Chip-loss regularisation
max(h, 0)
ψ_ε(h)
Effective thickness [mm]
Chip thickness h [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.0100 −0.0075 −0.0050 −0.0025 0.0000 0.0025 0.0050 0.0075 0.0100
(b) Geometric engagement and phase cells
g_i(θ)
θ_s
θ_e
Δθ
C_k (phase cell)
A_{i,r,k} = C_k ∩ E_{i,r}
active interval
C_k ∩ E_i = ∅
entry crossing
C_k ⊆ E_i (full)
exit crossing
C_k ∩ E_i = ∅
a = θ_s
b
a = θ_k, b = θ_k + Δθ
a
b = θ_e
i = 0
i = 1
i = 2
i = 3
p = 2π/N
n_e(θ)
∫_a^b F_i(ϑ) dϑ ≈ Σ_{j=1}^{n_q} F_i(ϑ_j) Δϑ
a_{i,r,k}
Δϑ
b_{i,r,k}
Active-interval quadrature in ϑ